

DEEPAK TITUS EDWARD

Staff Software Engineer -- Generative AI (RAG & Agentic Systems) | MarTech & Customer Data Platforms | AWS

Certified

San Francisco Bay Area | [linkedin.com/in/deepakedward](https://www.linkedin.com/in/deepakedward) | deepakedward.github.io

github.com/deepakedward | deepakedward.medium.com

PROFESSIONAL SUMMARY

Staff-level software engineer with 18+ years building large-scale distributed systems, real-time data pipelines, and cloud-native platforms -- most recently as a Generative-AI / RAG engineer and MarTech data-engineering lead at McAfee. Progressed from owning Adobe Experience Platform (AEP) / Customer Data Platform (CDP) engineering to owning end-to-end GenAI: architected and shipped an AWS Bedrock RAG chatbot and a custom LLM-as-a-judge evaluation framework that drove answer correctness from 0.17 to 0.83, and designed a 5-agent, LangGraph-based Next-Best-Action marketing orchestrator with a three-layer safety stack. In parallel, led a Tealium first-party-domain migration (2-5B hits/year) and a custom web-events streaming pipeline (15M events/day). Focused on end-to-end ownership, cross-functional alignment, and measurable production outcomes.

CORE SKILLS

Generative AI / Agentic: AWS Bedrock (Agents, Knowledge Bases, Guardrails, RAG Eval) - Amazon Nova - Anthropic Claude - Titan v2 Embeddings - Cohere Rerank - OpenSearch Serverless - Aurora pgvector - Amazon Lex - LangGraph - Databricks Mosaic AI & Genie - RAG, semantic caching, LLM-as-a-judge, prompt engineering, human-in-the-loop

Data & Cloud: Adobe Experience Platform / RT-CDP - Adobe Journey Optimizer - AEP Query Service / Data Distiller - Tealium iQ / EventStream - Akamai - LiveRamp - HighTouch - Braze - Kafka / Confluent - Databricks / Spark Structured Streaming - Delta Lake / Unity Catalog - AWS (Lambda, Step Functions, EventBridge, SQS, DynamoDB, ECS Fargate, S3, CloudWatch) - Kong API Gateway - SQL Server

Languages & Tooling: Python (pandas, scikit-learn, XGBoost, reportlab, psycpg2) - SQL - JavaScript (ES5) - Terraform - GitHub Actions / Harness CI/CD - Databricks Asset Bundles - Jest - Playwright - Postman - Git

Machine Learning: scikit-learn, XGBoost, SMOTE, hyperparameter tuning, ROC-AUC / F1 evaluation, model serving

KEY ACHIEVEMENTS

- Architected and shipped an end-to-end AWS Bedrock RAG chatbot on Amazon Nova + Titan v2 embeddings, with a 68-document multi-locale knowledge base, guardrails, and least-privilege IAM.
- Built a RAG evaluation framework (LLM-as-a-judge) over ~500-prompt datasets across 5 iterations -- drove answer correctness 0.17 -> 0.83, citation coverage 0.88, harmfulness 0.00.
- Designed a 5-agent Next-Best-Action marketing orchestrator on Claude with LangGraph durable orchestration, structured-output contracts, and a 3-layer safety stack (LLM critic + deterministic guardrails + human-in-the-loop); runs identically offline, on AWS Bedrock, and Databricks-native.
- Led a Tealium first-party-domain migration (Akamai reverse proxy) coordinating 6 teams across 2-5B hits/year; authored the Adobe Analytics impact-measurement framework and a 5-phase runbook.
- Built a custom web-events streaming CDP pipeline at 15M events/day / 500K-min peak with PII/non-PII segregation, idempotent Spark Structured Streaming, and a ~13 KB JS collection SDK.
- Modeled AEP/CDP data at 100M+ device scale; led a device-schema V1->V2 migration; built activation audiences of 60M+ profiles across 8+ AdTech destinations.
- Authored McAfee's Senior MarTech AI Engineer hiring framework and ran a 5-round interview program across ~10 candidates on two teams.

EXPERIENCE

McAfee -- Staff Software Engineer

July 2021 - Present | Santa Clara, CA

Leading MarTech data engineering and Generative-AI development for McAfee's Consumer business at 100M+ device scale on AWS and Adobe Experience Platform.

Generative AI / RAG & Agentic Systems (2025-2026)

- Built the full AWS Bedrock Agent + Knowledge Base stack end-to-end (guardrails, action groups, least-privilege IAM); stood up dual vector stores (OpenSearch Serverless + Aurora pgvector) to compare chunking strategies.
- Authored a production runbook with SLOs (99.9% availability, P95 < 2s), CloudWatch per-model token-cost dashboards, and P1 alarm playbooks.
- Engineered a verified semantic cache (>80% direct / 60-80% few-shot / <60% fallback) with reranking to cut latency, cost, and hallucinations; shipped a hallucination fix root-caused to KB source data (2 merged PRs).

- Designed the NBA Campaign Orchestrator -- a 5-agent fan-out/synthesize/critique system (~6,000 lines of Python) with deterministic guardrails, idempotency/frequency-capping, cohort aggregation, and a Databricks-native deployment (Unity Catalog -> Mosaic AI Model Serving -> Braze -> Streamlit HITL console).

MarTech Platform Engineering (2026)

- Led the Tealium first-party-domain migration to an Akamai reverse-proxy architecture; authored a beacon-level Adobe Analytics measurement plan and a full tag-validation suite.
- Designed a custom web-events streaming pipeline (Tealium -> JS SDK -> Kong -> Kafka -> Databricks medallion -> HighTouch) with dual PII/non-PII topics, two Avro schemas, CDF-driven idempotent MERGE dedup, and bot filtering; owned the production runbook.

Adobe Experience Platform / CDP, Privacy & ML (2021-2025)

- Identified and escalated a critical identity-stitching defect collapsing distinct users onto one profile; authored Query Service SQL to root-cause and verify the fix.
- Integrated 8+ AdTech destinations (LiveRamp, Meta, Yahoo, Rokt, PebblePost, Optizmo, Dynata, Google Ads) with SFTP/API onboarding and SHA-256 hashing; shipped LiveRamp eCST anonymous-user activation.
- Built a scikit-learn/XGBoost dynamic-pricing classifier (SMOTE, ROC-AUC); implemented GDPR pipelines (SHA-256) and 30+ CloudWatch dashboards with anomaly detection; migrated AEP auth JWT -> OAuth 2.0; ran zero-downtime Node.js Lambda upgrades across all envs.
- Automated AEP anonymous-profile cleanup, cutting ~2.5 hrs/day of manual work to zero; led the Journey Observability funnel release (Databricks -> MySQL, 10 dimensions).

Leadership

- Authored the Senior MarTech AI Engineer role definition and HR hiring guideline; designed a 5-round interview framework, weighted scorecard, and reportlab-generated question banks; served as interviewer/evaluator across ~10 candidates. Honored with "Outstanding Performance and Excellence in Execution."

HCL Technologies -- Senior Technical Architect

July 2020 - July 2021 | San Francisco Bay Area

- Built business-critical microservices (Spring Boot, Spring Batch) on Kubernetes and scalable data-ingestion tooling; reviewed architecture and partnered with product stakeholders on mission-critical platform tools.

realtor.com -- Staff Software Engineer (Lead Monetization)

August 2018 - June 2020 | San Francisco Bay Area

- Built large-scale distributed REST microservices (Spring Boot, AWS); migrated AWS Lambda services to an ECS Spring Batch application; built Splunk/New Relic observability; mentored engineers and set engineering best practices.

Infosys -- Technology Lead

April 2015 - July 2018 | San Francisco Bay Area

- Lead Java backend engineer for Apple's product-manufacturing team; designed scalable multithreaded REST APIs in a TDD environment; led front-end and back-end developers.

ITC Infotech -- Lead / Senior Software Engineer

July 2007 - April 2015 | SF Bay Area & Bengaluru, India

- Designed REST APIs on Google Cloud Platform for Motorola Mobility; core member of the Big Data Center of Excellence (Cloudera/Hortonworks); module lead for a UK online-banking portal and the Starwood Hotels reservation system.

EDUCATION & CERTIFICATIONS

- The University of Texas at Austin (McCombs) -- Post Graduate Program in AI & ML (2023-2024)
- MCA, Computer Science -- Dr. A.P.J. Abdul Kalam Technical University (2004-2007)
- B.Sc., Mathematics, Chemistry & Physics -- Kanpur University (2001-2004)
- Certifications: AWS Certified AI Practitioner (2025) - Databricks Certified GenAI Engineer Associate - AWS Solutions Architect Associate (prep) - Machine Learning Foundations - Introduction to Generative AI - Generative AI for Practitioners